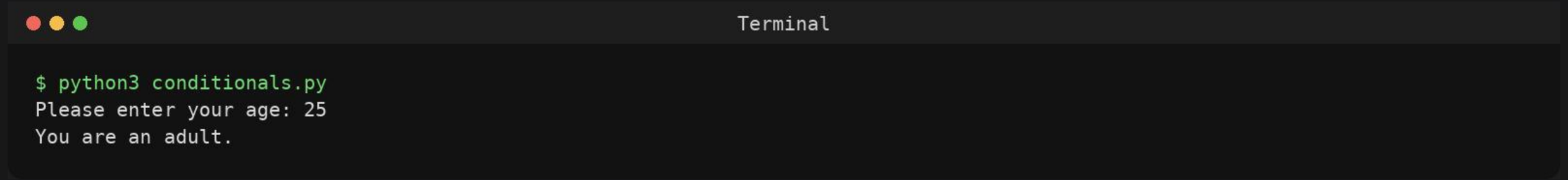
Master Class — Python
Code & Terminal Output Screenshots

23 files | code screenshot + terminal output for each

conditionals.py

```python
age = int(input("Please enter your age: "))
if age < 10:
    print("You are a minor.")
elif age < 18:
    print("You are a teenager.")
else:
    print("You are an adult.")
```


Terminal
$ python3 conditionals.py
Please enter your age: 25
You are an adult.

Data_str_Stud_mark_system.py

```python
# Student marks system
students = {}
st = """
options:
Enter 1 to Add student
Enter 2 for view records
"""
def add_student():
    name = input("Enter student Name: ")
    marks = input("Enter marks")
    students[name] = marks

def show_records():
    for s,m in students.items():
        print(f'{s} : {m}')


while True:
    print(st)
    choice = int(input("Enter number: "))

    if choice ==1 :
       add_student()
    elif(choice == 2):
        show_records()

```

Terminal

```
$ python3 Data_str_Stud_mark_system.py

options:
Enter 1 to Add student
Enter 2 for view records

Enter number: 1
Enter student Name: Ali
Enter marks85

options:
Enter 1 to Add student
Enter 2 for view records

Enter number: 2
Ali : 85

options:
Enter 1 to Add student
Enter 2 for view records

Enter number:

[... program is still running, waiting for the next input ...]
```

Dictionaries.py

```python
dic = {
    "name": "Noman",
    "age" : 19,
    "semester": 3
}
print(dic)
print(dic["name"])
# adding new key and value in dictionary
dic["Department"] = "CS and IT"
print(dic)
# Dic are mutable
dic["age"] = 20
print(dic)
# Checking existence

print("name" in dic)
print("name" not in dic)

del dic["Department"]
print(dic)
print(dic.items())
print(dic.keys())
print(dic.values())
```

Terminal

```
$ python3 Dictionaries.py
{'name': 'Noman', 'age': 19, 'semester': 3}
Noman
{'name': 'Noman', 'age': 19, 'semester': 3, 'Department': 'CS and IT'}
{'name': 'Noman', 'age': 20, 'semester': 3, 'Department': 'CS and IT'}
True
False
{'name': 'Noman', 'age': 20, 'semester': 3}
dict_items([('name', 'Noman'), ('age', 20), ('semester', 3)])
dict_keys(['name', 'age', 'semester'])
dict_values(['Noman', 20, 3])
```

Error_Handling.py

```python
try:
    num = int(input("Enter a number: "))
    result = 10 / num
    print("Result is:", result)
except ValueError:
    print("Error: Invalid input. Please enter a valid integer.")
except ZeroDivisionError:
    print("Error: Division by zero is not allowed.")
```

Terminal

```
$ python3 Error_Handling.py
Enter a number: 5
Result is: 2.0
```

Exception_handling.py

```python
def log_error(error):
    with open('file_handling_files/error_log.txt', 'a') as file:
       file.write(f"{error}\n")

def safe_cal():
    try:
        num1 = float(input("Enter first number: "))
        operator = input("Enter operator (+, -, *, /): ")
        num2 = float(input("Enter second number: "))
        if operator == '+':
            result = num1 + num2
        elif operator == '-':
            result = num1 - num2
        elif operator == '*':
            result = num1 * num2
        elif operator == '/':
            result = num1 / num2
            if num2 == 0:
                raise ZeroDivisionError("Cannot divide by zero")
        else:
            raise ValueError("Invalid operator")
        print(f"Result: {result}")
    except Exception as e:
        print(f"error : {e}")
        log_error(e)

    finally:
        print("execution completed.")

while True:
    print("----safe calculator----")
    safe_cal()
    again = input("Do you want to perform another calculation? (yes/no): ")
    if again.lower() != 'yes':
        print("Goodbye!")
        break
```

Terminal

```
$ python3 Exception_handling.py
----safe calculator----
Enter first number: 10
Enter operator (+, -, *, /): +
Enter second number: 5
Result: 15.0
execution completed.
Do you want to perform another calculation? (yes/no): no
Goodbye!
```

File_Handling.py

```python
#  writing
with open("file_handling_files/students.txt", "w") as stu_file:
    w_stu = stu_file.write("This text is write by .write mode")
    print(w_stu)


with open("file_handling_files/data.txt", "w") as data_file:
    data_file.write("This text is write by .write mode")
    print("Data file is created and text is written in it")

with open("file_handling_files/students.txt", "r") as stu_file:
    con_stu = stu_file.read()


with open("file_handling_files/data.txt", "r") as data_file:
    con_data = data_file.read()
    print(con_data)

    print(con_stu)
```

Terminal

```
$ python3 File_Handling.py
33
Data file is created and text is written in it
This text is write by .write mode
This text is write by .write mode
```

Funtions.py

```python
def sqr(n):
    return n*n
def bigger(n,m):
    if n > m:
        return n
    elif m > n:
        return m

print(sqr(5))
print(bigger(2,3))

# args and kwargs

def add_numbers(*args):
    return sum(args)

def student_system(**kwargs):
    for key, value in kwargs.items():
        print(f"{key} : {value}")

add_numbers(1,2,3,4,5)
student_system(name= "Noman",age = 20)
```

Terminal

```
$ python3 Funtions.py
25
3
name : Noman
age : 20
```

List.py

```python
lst = [3,4,5,2,5,5,6,9,4,2,5,11]
lst.append(45)
print(lst)
print(lst[0])
print(lst[-1])
print(lst[1:3])

lst[4] = 44
print(lst)

# --
print("List Method")
print(lst)
lst.insert(1,4)  # for inserting item at specific index
lst.remove(5)  # for removing item form list
print(lst.pop())   # for removing last item form list
print(lst.index(4))
```

Terminal

```
$ python3 List.py
[3, 4, 5, 2, 5, 5, 6, 9, 4, 2, 5, 11, 45]
3
45
[4, 5]
[3, 4, 5, 2, 44, 5, 6, 9, 4, 2, 5, 11, 45]
List Method
[3, 4, 5, 2, 44, 5, 6, 9, 4, 2, 5, 11, 45]
45
1
```

Loops.py

```python
# use of continue statement

print("Continue statement example:")
for i in range(5):
    if i == 3:
        continue
    print(i)

# use of break statement
print("Break statement example:")
count = 0
while count < 5:
    if count == 3:
        break
    print(count)
    count += 1

# nested loops
print("Nested loops example:")
for i in range(3):
    for j in range(3):
        print(f"({i}, {j})")

#  Multiplication table using  loops
print("Multiplication table:")
input_number = int(input("Enter a number: "))
for i in range(1, 11):
    print(f"{input_number} x {i} = {input_number * i}")
```

Terminal

```
$ python3 Loops.py
Continue statement example:
0
1
2
4
Break statement example:
0
1
2
Nested loops example:
(0, 0)
(0, 1)
(0, 2)
(1, 0)
(1, 1)
(1, 2)
(2, 0)
(2, 1)
(2, 2)
Multiplication table:
Enter a number: 7
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70
```

Loops_hands_on_project.py

```python
class MultiplicationTable:
    def __init__(self, number):
        self.number = number

    def display_table(self):
        for i in range(1, 11):
            print(f"{self.number} x {i} = {self.number * i}")

m1 = MultiplicationTable(5)
m1.display_table()
```

Terminal

```
$ python3 Loops_hands_on_project.py
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

MyModule.py

```
def add(n,m):
    print(n+m)
def sub(n,m):
    print(n-m)
def sqr(n):
    print(n*n)
```

Terminal

```
$ python3 MyModule.py
```

funcations_and_Modules.py

```python
import MyModule as mm

mm.add(5,4)
mm.sqr(5)
mm.sub(6,5)
```

Terminal

```
$ python3 funcations_and_Modules.py
9
25
1
```

inheritance_&_polymorphism.py

```python
class Student:

    def __init__(self, name, age, class_name, student_id):
        self.name = name
        self.age = age
        self.class_name = class_name
        self.__student_id = student_id

    def set_student_id(self, new_id):
        self.__student_id = new_id

    def get_student_id(self):
        return self.__student_id

    def get_info(self):
        print(
            {
                "Name": self.name,
                "Age": self.age,
                "Class": self.class_name,
                "Student ID": self.__student_id,
            }
        )

class Teacher(Student):
    def __init__(self, name, age, class_name, student_id, subject):
        Student.__init__(self, name, age, class_name, student_id)
        self.subject = subject

    def get_info(self):
        print(
            {
                "Name": self.name,
                "Age": self.age,
                "Department": self.class_name,
                "Teacher ID": self.get_student_id(),
                "Subject": self.subject,
            }
        )

s1 = Student("Salman", 10, "8th", 4)
s1.get_info()

t1 = Teacher("Noman", 20, "BSAI", 50, "Python")
t1.set_student_id(20)
t1.get_info()
```

Terminal

```
$ python3 inheritance_&_polymorphism.py
{'Name': 'Salman', 'Age': 10, 'Class': '8th', 'Student ID': 4}
{'Name': 'Noman', 'Age': 20, 'Department': 'BSAI', 'Teacher ID': 20, 'Subject': 'Python'}
```

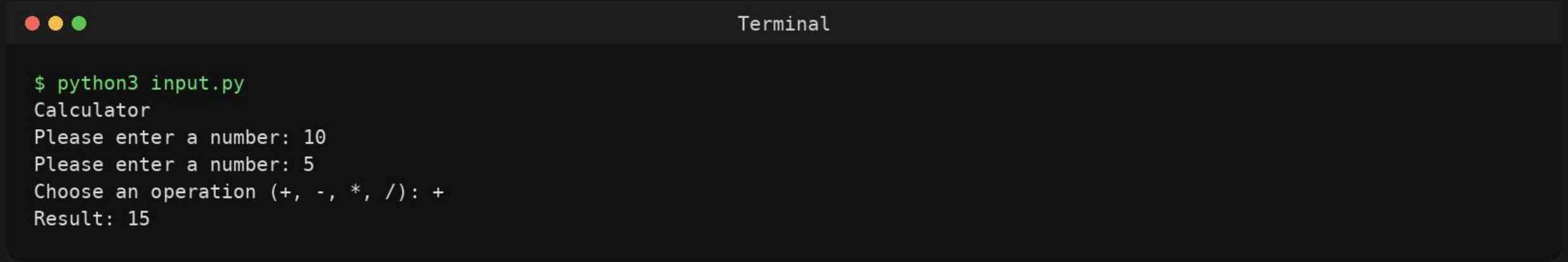
input.py

```python
print("Calculator")


value_1 = int(input("Please enter a number: "))
value_2 = int(input("Please enter a number: "))

choice = input("Choose an operation (+, -, *, /): ")
if choice == '+':
    result = value_1 + value_2
elif choice == '-':
    result = value_1 - value_2
elif choice == '*':
    result = value_1 * value_2
elif choice == '/':
    result = value_1 / value_2
else:
    result = "Invalid operation"

print(f"Result: {result}")
```


Terminal
$ python3 input.py
Calculator
Please enter a number: 10
Please enter a number: 5
Choose an operation (+, -, *, /): +
Result: 15

masterclass.py

```python
# Student managment system
class Student:
    def __init__(self,name,student_id,age):

        self.name = name
        self.student_id = student_id
        self.age = age

    def show(self):
        print(f"""
ID : {self.student_id}
Name : {self.name}
Age : {self.age}""")

students = []
# Adding new Studnet to list
def add_student():
    try:
        student_id = int(input("Enter Student id: "))
        name = input("Enter Student name: ")
        age = int(input("Enter Student age: "))

        student = Student(name, student_id, age)

        students.append(student)
        print("Student added")
    except ValueError:
        print("Invalid input. Please enter valid data.")

# Showing Studnets
def view_students():
    if not students:
        print("No student found")
    for student in students:
        student.show()

def search_student():
    try:
        search_id = int(input("Enter student id to search: "))
        found = False
        for student in students:
            if student.student_id == search_id:
                student.show()
                found = True
                break
        if not found:
            print("Student not found")
    except ValueError:
        print("Enter valid search id")


while True:
    print(f"""
    1. Add Student
    2. View Students
    3. search studnet
    4. exit """)
    try:
        choice = int(input("Enter choice: "))
    except ValueError:
        print("Invalid input. Please enter a number.")

    if choice == 1:
        add_student()
    elif choice == 2:
        view_students()
    elif choice == 3:
        search_student()
    elif choice == 4:
        print("Exiting")
        break
    else:
        print("Invalid choice. Please try again.")
```

Terminal

```
$ python3 masterclass.py

    1. Add Student
    2. View Students
    3. search studnet
    4. exit
Enter choice: 1
Enter Student id: 1
Enter Student name: Ali
Enter Student age: 20
Student added

    1. Add Student
    2. View Students
    3. search studnet
    4. exit
Enter choice: 3
Enter student id to search: 1

ID : 1
Name : Ali
Age : 20

    1. Add Student
    2. View Students
    3. search studnet
    4. exit
Enter choice: 4
Exiting
```

object_oriented_programming.py

```python
class Computer:
    def __init__(self, brand, model, processor, ram, storage):
        self.brand = brand
        self.model = model
        self.processor = processor
        self.ram = ram
        self.storage = storage

    def display_specs(self):
        print(f"Brand: {self.brand}")
        print(f"Model: {self.model}")
        print(f"Processor: {self.processor}")
        print(f"RAM: {self.ram} GB")
        print(f"Storage: {self.storage} GB")

c1 = Computer("hp", " gen","i5", 8, 256)
c1.display_specs()
```

Terminal

```
$ python3 object_oriented_programming.py
Brand: hp
Model:  gen
Processor: i5
RAM: 8 GB
Storage: 256 GB
```

sets.py

```python
s1 = {1,2,3,4,5,6,7,8,9}
s2 = {9,10,11,12,13,14,15}
print(s1.union(s2))
print(s1.intersection(s2))
print(s1.difference(s2))
print(s1.symmetric_difference(s2))
```

Terminal

```
$ python3 sets.py
{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}
{9}
{1, 2, 3, 4, 5, 6, 7, 8}
{1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15}
```

student_data_management_system.py

```python
import json
import os
path = 'file_handling_files/student.json'

def add_student():
    name  = input("Enter Name: ")
    age = int(input("Enter age: "))
    class_name = input("Enter Class name: ")

    new_student = {
        "name" : name,
        "age"  : age,
        "class_name" : class_name
        }

    if os.path.exists(path) and os.path.getsize(path) > 0:
        with open(path, "r") as file:
            students = json.load(file)
    else:
        students = []

    students.append(new_student)

    with open(path, "w") as file:
        json.dump(students, file, indent=4)


def view_students():
    if os.path.exists(path) and os.path.getsize(path) > 0:
        with open(path, "r") as file:
            data = json.load(file)
            print(data)
    else:
        print("No student data found.")

while True:
    print("1. Add Student")
    print("2. View Students")
    print("3. Exit")
    choice = int(input("Enter Choice: "))

    if choice == 1:
        add_student()
    if choice == 2:
          view_students()
    if choice == 3:
          break
```

Terminal

```
$ python3 student_data_management_system.py
1. Add Student
2. View Students
3. Exit
Enter Choice: 1
Enter Name: Ali
Enter age: 20
Enter Class name: BSCS
1. Add Student
2. View Students
3. Exit
Enter Choice: 2
[{'name': 'Salman', 'age': 25, 'class_name': 'ics'}, {'name': 'noman', 'age': 20, 'class_name': 'bsai'}, {'name':
'hanza', 'age': 15, 'class_name': 'bscs'}, {'name': 'Ali', 'age': 20, 'class_name': 'BSCS'}]
1. Add Student
2. View Students
3. Exit
Enter Choice: 3
```

working_with_Formatted_user_profile_System.py

```python
class CreateUserProfile:
    def __init__(self, id, name, email, age):
        self.id = id
        self.name = name.title()
        self.email = email.lower()
        self.age = age.strip()

        profile_data = f"""
ID: {self.id}
Name: {self.name}
Email: {self.email}
Age: {self.age}
 -------------------

 """
        with open("file_handling_files/user_profile.txt", "a") as file:
            file.write(profile_data)
        print("User profile created successfully! \n")

user_if = 1

while True:
    name = input("Enter your name: ")
    email = input("Enter your email: ")
    age = input("Enter your age: ")
    user_profile = CreateUserProfile(user_if, name, email, age)

    user_if += 1

    choice = input("Do you want to create another profile? (yes/no): ")
    if choice.lower() != "yes":
        print("Goodbye!")
        break
```

Terminal

```
$ python3 working_with_Formatted_user_profile_System.py
Enter your name: Ali Khan
Enter your email: ali@gmail.com
Enter your age: 20
User profile created successfully!

Do you want to create another profile? (yes/no): no
Goodbye!
```

working_with_String.py

```python
s1 = "This is a string"
s2 = "This is another string"
print(len(s1))
print(s1.upper())
print(s1.lower())
print(s1[0:4])

s3 = s1 + " " + s2
print(s3)
```

Terminal

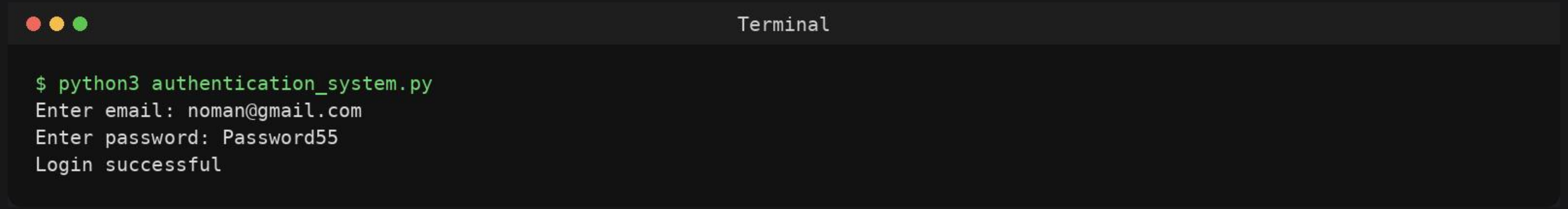
```
$ python3 working_with_String.py
16
THIS IS A STRING
this is a string
This
This is a string This is another string
```

authentication_system.py

```python
email = input("Enter email: ")
password = input("Enter password: ")

with open("students.txt","r") as file:
    for line in file:
        s_name, s_email, s_password = line.strip().split(",")

        if email == s_email and password == s_password:
            print("Login successful")
            break
        else:
            print("Login fail")
```


Terminal
$ python3 authentication_system.py
Enter email: noman@gmail.com
Enter password: Password55
Login successful

number_guess_game.py

```python
import random
number = random.randint(1, 100)
count = 0
while True:
    guess = int(input("Guess a number between 1 and 100: "))
    count += 1
    if guess < number:
        print("Too low! Try again.")
    elif guess > number:
        print("Too high! guess again")
    elif guess == number:
        print("You win the game")
        break
    else:
        print("Plese Enter a valid input")

print(f"Attempts : {count}")
```

Terminal

```
$ python3 number_guess_game.py
Guess a number between 1 and 100: 50
Too low! Try again.
Guess a number between 1 and 100: 75
Too low! Try again.
Guess a number between 1 and 100: 88
Too high! guess again
Guess a number between 1 and 100: 81
Too low! Try again.
Guess a number between 1 and 100: 84
Too high! guess again
Guess a number between 1 and 100: 82
You win the game
Attempts : 6
```

registration_system.py

```python
print("Student Registration")

while True:
    name = input("Enter name: ")
    email = input("Enter email: ")
    while "@" not in email or "." not in email or " " in email:
        email = input("Please enter a valid email: ")
    password = input("Enter password: ")
    while len(password) <8 or not any(char.isupper() for char in password) or not any(char.isdigit() for char in password):
        password = input("Enter a valid password: ")

    with open("students.txt", "a") as file:
        file.write(f"{name},{email},{password}\n")
    print("Student added in file")
```


Terminal
$ python3 registration_system.py
Student Registration
Enter name: Bilal Ahmed
Enter email: bilal@example.com
Enter password: Bilal2025
Student added in file
Enter name:
[... program is still running, waiting for the next input ...]